

Quest: Frameworks in Stochastic Calculus, Numerical Discretization, and Multi-Domain Stochastic Modeling

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Abstract

This comprehensive report investigates the mathematical foundations, numerical approximation techniques, and practical applications of continuous-time stochastic processes. Moving systematically from foundational probability to applied continuous systems, we first define the axiomatic properties of Standard Brownian Motion and establish its macroscopic universality via Donsker's Invariance Principle. We provide a rigorous calculus framework by evaluating Itô's integral formulations and deriving Itô's first and second lemmas via Taylor expansions and quadratic variations. To bridge continuous-time theory with computational architecture, we analyze the Euler-Maruyama numerical discretization scheme, evaluating its strong and weak convergence criteria and implementing log-transformations to resolve non-physical boundary behavior. These frameworks are deployed across three critical modeling paradigms: asset price dynamics under the Black-Scholes geometric framework alongside Monte Carlo option pricing; mean-reverting path configurations governed by the Ornstein-Uhlenbeck process; and discrete birth-death queuing dynamics applied to structural steady-state optimizations.

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1 Introduction

Many systems across physics, economics, and operations research are subject to unavoidable fluctuations, making deterministic models insufficient. Stochastic differential equations (SDEs) solve this by combining deterministic trends with random processes, providing a rigorous way to analyze paths that are continuous everywhere but differentiable nowhere.

This project, titled **Quest**, builds a rigorous mathematical and computational bridge between theoretical stochastic calculus and practical multi-domain modeling. Rather than analyzing these topics in isolation, this paper integrates continuous mathematical abstractions with algorithmic verification.

The structural trajectory of this report is organized as follows:

- **Stochastic Foundations:** Building a rigorous baseline for Wiener processes, analyzing low-order statistical moments, and verifying macroscopic convergence behaviors.
- **Analytical Frameworks:** Deriving Itô's Lemma to establish a functional calculus capable of handling the non-zero quadratic variation inherent to random perturbations.
- **Algorithmic Discretization:** Deploying the Euler-Maruyama approximation, evaluating path tracking stability, and solving positivity constraints.
- **Applied Modeling Paradigms:** Implementing these mathematical toolsets to solve three classical problems: the geometric Black-Scholes option pricing model, mean-reverting Ornstein-Uhlenbeck structures, and M/M/1 queuing behavior under steady-state assumptions.

By unifying abstract probability theory with computational implementation, this report demonstrates how modern stochastic calculus serves as a definitive language for tracking uncertainty in complex systems.

2 Foundations of Stochastic Processes and Brownian Motion

2.1 Formal Axiomatic Definition of a Wiener Process

A continuous-time stochastic process $\{W(t)\}_{t \geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is classified as a Standard Brownian Motion, or a Wiener process, if it strictly satisfies the following four axiomatic conditions:

1. **Initial Condition:** The process starts deterministically at the origin with absolute certainty:

$$W(0) = 0 \quad \text{with probability } 1 \tag{1}$$

2. **Continuity of Paths:** The sample path mapping $t \mapsto W(t, \omega)$ is a continuous function of time t for almost all realizations $\omega \in \Omega$.

3. **Independent Increments:** For any finite sequence of ordered timestamps $0 \leq t_1 < t_2 < \dots < t_n < \infty$, the corresponding forward-looking variations:

$$W(t_2) - W(t_1), \quad W(t_3) - W(t_2), \quad \dots, \quad W(t_n) - W(t_{n-1}) \tag{2}$$

are mutually independent random variables. This property ensures that the future evolution of the trajectory is uninfluenced by its historical path.

4. **Stationary Gaussian Increments:** For any two timestamps $0 \leq s < t$, the incremental change $W(t) - W(s)$ is a normally distributed random variable with mean zero and a variance scaling linearly with the elapsed time duration:

$$W(t) - W(s) \sim \mathcal{N}(0, t - s) \tag{3}$$

The probability density function $f(x, t)$ tracking the absolute position of $W(t)$ given $W(0) = 0$ is derived directly from this Gaussian increment profile by setting $s = 0$:

$$f(x, t) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right) \quad (4)$$

2.2 Statistical Moments and Covariance Properties

We evaluate the low-order structural characteristics that dictate the trajectory bounds of a standard Wiener process. Because $W(t) \sim \mathcal{N}(0, t)$, the expectation operator directly yields the first algebraic moment:

$$\mathbb{E}[W(t)] = 0, \quad \forall t \geq 0 \quad (5)$$

The second central moment represents the variance, which grows linearly over time due to the dispersion of potential sample paths:

$$\text{Var}(W(t)) = \mathbb{E}[W(t)^2] - (\mathbb{E}[W(t)])^2 = t \quad (6)$$

To uncover the temporal dependence structure between different positions along the same path, we derive the autocovariance function $\text{Cov}(W(s), W(t))$ for arbitrary indices $s, t \geq 0$. Without loss of generality, let us assume an ordered temporal reference frame such that $s \leq t$. By adding and subtracting the intermediate state $W(s)$, we can isolate an independent increment segment:

$$\text{Cov}(W(s), W(t)) = \mathbb{E}[W(s)W(t)] - \mathbb{E}[W(s)]\mathbb{E}[W(t)] \quad (7)$$

$$= \mathbb{E}[W(s)(W(t) - W(s) + W(s))] - 0 \quad (8)$$

$$= \mathbb{E}[W(s)(W(t) - W(s))] + \mathbb{E}[W(s)^2] \quad (9)$$

Exploiting the independent increments axiom, the joint expectation of the product splits into the product of individual expectations:

$$\mathbb{E}[W(s)(W(t) - W(s))] = \mathbb{E}[W(s)] \cdot \mathbb{E}[W(t) - W(s)] = 0 \cdot 0 = 0 \quad (10)$$

Substituting this back into the relation leaves only the second moment of the earlier step:

$$\text{Cov}(W(s), W(t)) = 0 + \mathbb{E}[W(s)^2] = s \quad (11)$$

Generalizing this derivation to remove the $s \leq t$ ordering assumption shows that the autocovariance is strictly bounded by the minimum of the two time coordinates:

$$\text{Cov}(W(s), W(t)) = \min(s, t) \quad (12)$$

2.3 Macroscopic Universality via Donsker's Invariance Principle

The significance of Brownian Motion extends beyond its axiomatic construction; it arises naturally as the macroscopic scaling limit of discrete stochastic actions. This behavior is formally codified by Donsker's Invariance Principle, also known as the functional central limit theorem.

Let $\{X_i\}_{i=1}^{\infty}$ represent an infinite sequence of independent, identically distributed (i.i.d.) random variables centered at a zero mean ($\mathbb{E}[X_i] = 0$) with a finite variance ($\text{Var}(X_i) = \sigma^2$). We build a standard discrete random walk S_n by taking successive cumulative sums:

$$S_n = \sum_{i=1}^n X_i \quad (13)$$

To map this discrete process onto a continuous time domain $t \in [0, 1]$, we define a rescaled continuous-time stochastic process $W_n(t)$ by applying a floor function $\lfloor nt \rfloor$ to sample the random walk index:

$$W_n(t) = \frac{1}{\sigma\sqrt{n}} S_{\lfloor nt \rfloor} \quad (14)$$

Donsker's Invariance Principle states that as the discretization resolution approaches infinity ($n \rightarrow \infty$), the entire sequence of functions $W_n(t)$ converges weakly in distribution to a standard Wiener process $W(t)$:

$$W_n(t) \xrightarrow{d} W(t) \quad \text{as } n \rightarrow \infty \quad (15)$$

This indicates that independent of the underlying microscopic step distribution (whether discrete Bernoulli jumps or continuous distributions), the scaled multi-step path profiles always converge toward continuous Gaussian Brownian paths.

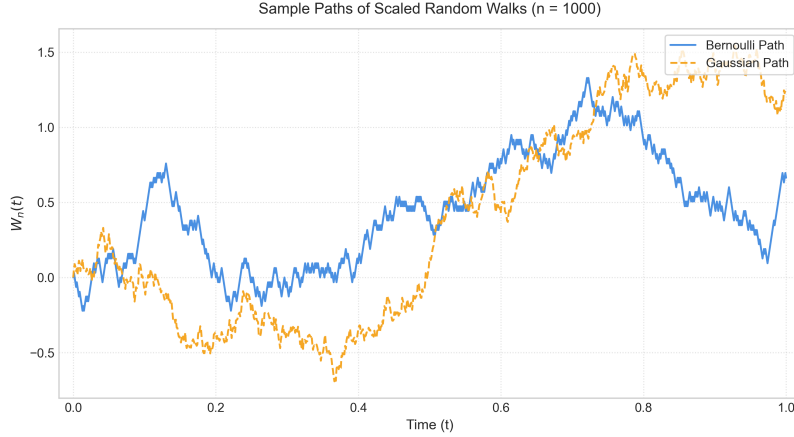


Figure 1: Visualizing path convergence under Donsker's Invariance Principle. As the step count reaches $n = 1000$, discrete steps become macroscopically indistinguishable from a true continuous Wiener process.

3 Stochastic Calculus and the Derivation of Itô's Lemma

3.1 The Itô Stochastic Integral and Simple Processes

Classical Riemann-Stieltjes calculus fails when applied to trajectories driven by Brownian Motion because a standard Wiener path $W(t)$ possesses infinite total variation over any finite interval $[0, T]$. To establish a mathematically rigorous integration framework, we define the Itô stochastic integral for a class of adapted, non-anticipating integrand processes $A(t)$.

We first define this operation for a *simple process* (or step process), where the time domain is partitioned into discrete intervals $0 = t_0 < t_1 < \dots < t_m = T$, and the value of $A(t)$ remains constant as a random variable Y_i within each left-closed panel $t \in [t_i, t_{i+1})$. For a simple process, the Itô stochastic integral is explicitly defined as a discrete summation of increments:

$$\int_0^T A(s) dW(s) = \sum_{i=0}^{m-1} Y_i [W(t_{i+1}) - W(t_i)] \quad (16)$$

By extending this construction via a limiting sequence of simple approximations under the $\mathcal{L}^2$ norm, the continuous Itô stochastic integral is established for all square-integrable adapted processes.

3.2 Quadratic Variation of Brownian Motion

The defining characteristic that differentiates stochastic calculus from ordinary calculus is the concept of non-zero quadratic variation. Let $\Pi = \{t_0, t_1, \dots, t_n\}$ denote a partition of the time interval $[0, t]$, and let $\|\Pi\| = \max_i(t_{i+1} - t_i)$ represent the mesh size. The quadratic variation

of a standard Wiener process, denoted as $[W, W]_t$, is defined as the limit of the sum of squared increments:

$$[W, W]_t = \lim_{\|\Pi\| \rightarrow 0} \sum_{i=0}^{n-1} (W(t_{i+1}) - W(t_i))^2 \quad (17)$$

We evaluate this limit by checking its convergence in the mean-square sense ($\mathcal{L}^2$). Let $\Delta W_i = W(t_{i+1}) - W(t_i)$ and $\Delta t_i = t_{i+1} - t_i$. The expectation of the sum is:

$$\mathbb{E} \left[\sum_{i=0}^{n-1} (\Delta W_i)^2 \right] = \sum_{i=0}^{n-1} \mathbb{E}[(\Delta W_i)^2] = \sum_{i=0}^{n-1} \Delta t_i = t \quad (18)$$

By computing the variance of this sum and exploiting the fact that the kurtosis of a Gaussian distribution implies $\mathbb{E}[(\Delta W_i)^4] = 3(\Delta t_i)^2$, we find that the variance approaches zero as the mesh size shrinks to zero:

$$\lim_{\|\Pi\| \rightarrow 0} \text{Var} \left(\sum_{i=0}^{n-1} (\Delta W_i)^2 \right) = \lim_{\|\Pi\| \rightarrow 0} \sum_{i=0}^{n-1} 2(\Delta t_i)^2 \leq 2 \lim_{\|\Pi\| \rightarrow 0} \|\Pi\| \sum_{i=0}^{n-1} \Delta t_i = 2 \cdot 0 \cdot t = 0 \quad (19)$$

Because the mean-square limit has zero variance, the quadratic variation converges almost surely to a deterministic identity:

$$[W, W]_t = t \quad (\text{a.s.}) \quad (20)$$

In differential notation, this fundamental property is written as:

$$(dW)^2 = dt \quad (21)$$

Crucially, standard cross-products such as $dt \cdot dW$ and $(dt)^2$ grow at higher orders ($dt^{1.5}$ and dt^2), which drop out in the limit, evaluating identically to zero.

3.3 Derivation of Itô's First Lemma

Itô's first lemma provides an analytical framework to find the differential of a non-linear function dependent solely on a Wiener process, $f(W(t))$. To derive this, we partition the interval $[0, t]$ into n equal steps and decompose the net variation into a telescoping sum:

$$f(W(t)) - f(W(0)) = \sum_{j=1}^n [f(W(t_j)) - f(W(t_{j-1}))] \quad (22)$$

We expand each individual increment using a second-order Taylor series expansion about the left endpoint $W(t_{j-1})$:

$$f(W(t_j)) - f(W(t_{j-1})) = f'(W(t_{j-1}))\Delta W_j + \frac{1}{2}f''(W(t_{j-1}))(\Delta W_j)^2 + \mathcal{O}((\Delta W_j)^3) \quad (23)$$

Summing these expanded components across all sub-intervals yields:

$$f(W(t)) - f(W(0)) = \sum_{j=1}^n f'(W(t_{j-1}))\Delta W_j + \frac{1}{2} \sum_{j=1}^n f''(W(t_{j-1}))(\Delta W_j)^2 + \sum_{j=1}^n \mathcal{O}((\Delta W_j)^3) \quad (24)$$

Taking the continuous limit as $n \rightarrow \infty$ (and thus $\max \Delta t_j \rightarrow 0$), the first sum converges by definition to the Itô stochastic integral, the second sum converges to a deterministic time integral due to the quadratic variation identity $(\Delta W_j)^2 \rightarrow dt$, and the higher-order error terms vanish. This completes the proof of Itô's first lemma:

$$f(W(t)) = f(W(0)) + \int_0^t f'(W(s)) dW(s) + \frac{1}{2} \int_0^t f''(W(s)) ds \quad (25)$$

In compact differential format, this is expressed as:

$$df(W(t)) = f'(W(t)) dW(t) + \frac{1}{2} f''(W(t)) dt \quad (26)$$

3.4 Extension to Itô's Second Lemma

We extend this framework to a more general time-dependent stochastic process $X(t)$ governed by the standard stochastic differential equation:

$$dX(t) = a(t, X(t)) dt + b(t, X(t)) dW(t) \quad (27)$$

Consider a scalar function $f(t, X(t))$ that is continuously differentiable in time t and twice continuously differentiable in space X . Expanding $f(t, X(t))$ via a bivariate second-order Taylor series yields:

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial X} dX + \frac{1}{2} \frac{\partial^2 f}{\partial X^2} (dX)^2 + \frac{1}{2} \frac{\partial^2 f}{\partial t^2} (dt)^2 + \frac{\partial^2 f}{\partial t \partial X} dt dX \quad (28)$$

To evaluate the $(dX)^2$ operator, we square the explicit definition of our process:

$$(dX)^2 = (a dt + b dW)^2 = a^2 (dt)^2 + 2ab dt dW + b^2 (dW)^2 \quad (29)$$

Applying our multi-scale multiplication rules ($(dW)^2 = dt$, $(dt)^2 = 0$, $dt \cdot dW = 0$), the expression simplifies immediately to:

$$(dX)^2 = b^2 dt \quad (30)$$

Substituting $(dX)^2$ and the explicit formulation of dX back into the primary Taylor expansion, and dropping higher-order terms ($dt \cdot dX \rightarrow 0$, $(dt)^2 \rightarrow 0$), yields the final formulation for Itô's Second Lemma:

$$df(t, X(t)) = \left(\frac{\partial f}{\partial t} + a(t, X) \frac{\partial f}{\partial X} + \frac{1}{2} b(t, X)^2 \frac{\partial^2 f}{\partial X^2} \right) dt + b(t, X) \frac{\partial f}{\partial X} dW(t) \quad (31)$$

4 Numerical Discretization Schemes and Stability

4.1 The Euler-Maruyama Approximation Framework

Because most stochastic differential equations do not admit analytical closed-form solutions, numerical approximation techniques are necessary to simulate sample paths. Consider a general continuous-time stochastic differential equation of the form:

$$dX(t) = f(X(t)) dt + g(X(t)) dW(t), \quad X(0) = X_0, \quad 0 \leq t \leq T \quad (32)$$

where $f(X(t))$ represents the deterministic drift coefficient and $g(X(t))$ represents the stochastic diffusion coefficient.

To approximate this continuous process algorithmically, we select a uniform partition of the time domain $[0, T]$ into N equal steps of size $\Delta t = T/N$, where the discrete grid points are indexed as $t_n = n\Delta t$ for $n \in \{0, 1, \dots, N\}$. The Euler-Maruyama scheme acts as the stochastic extension of the classical deterministic Euler method, mapping the continuous differential equation onto a recursive discrete difference equation:

$$X_{n+1} = X_n + f(X_n)\Delta t + g(X_n)\Delta W_n \quad (33)$$

where the discrete Brownian motion increment ΔW_n is defined as the forward difference across the grid boundary:

$$\Delta W_n = W(t_{n+1}) - W(t_n) \quad (34)$$

Invoking the axiomatic properties of stationary Gaussian increments established in Section 2, each independent increment can be generated computationally by scaling a standard normal random variable:

$$\Delta W_n \approx \sqrt{\Delta t} \cdot Z_n, \quad \text{where } Z_n \sim \mathcal{N}(0, 1) \quad (35)$$

4.2 Convergence Criteria and Strong vs. Weak Classifications

To evaluate the mathematical validity of an approximation scheme over a refined grid, we assess its convergence properties as the step size approaches zero ($\Delta t \rightarrow 0$). Stochastic convergence splits into two distinct operational definitions based on whether we prioritize tracking specific individual paths or long-term statistical averages:

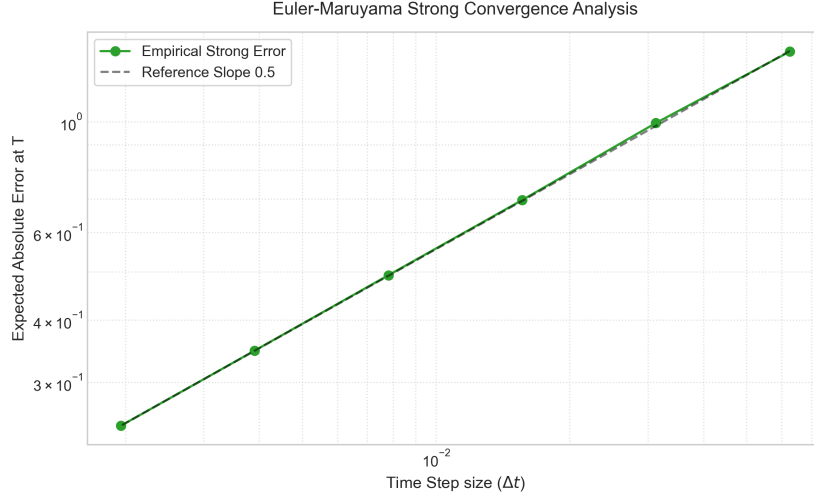


Figure 2: Log-log structural error profile tracking terminal variance differences. The empirical trace matches the standard 0.5 reference boundary, validating path-wise strong convergence bounds.

1. **Strong Convergence (Pathwise Tracking):** A numerical scheme exhibits strong convergence of order γ if the expected value of the absolute error between the true analytical path $X(T)$ and the approximate terminal state X_N is bounded by a linear function of the step size:

$$\mathbb{E} [|X(T) - X_N|] \leq C \cdot (\Delta t)^\gamma \quad (36)$$

where C is a generic positive constant independent of Δt . For the Euler-Maruyama method, the strong convergence order is $\gamma = 0.5$. This fractional order reflects the jagged, non-differentiable structure of the underlying Wiener paths, meaning that pathwise accuracy increases at a slower rate relative to deterministic schemes.

2. **Weak Convergence (Distributional Tracking):** A numerical scheme exhibits weak convergence of order β if the error between the expected values of any sufficiently smooth test function $\phi(X)$ under the true and simulated distributions scales with the step size:

$$|\mathbb{E}[\phi(X(T))] - \mathbb{E}[\phi(X_N)]| \leq C \cdot (\Delta t)^\beta \quad (37)$$

For the Euler-Maruyama method, the weak convergence order is $\beta = 1.0$. This higher order indicates that while individual sample paths can drift away from their true analytical trajectories, the collective distribution and its low-order moments (such as the mean and variance) converge linearly with the time step.

4.3 The Boundary Positivity Constraint and Log-Transformations

Many physical and financial quantities—such as asset prices, absolute temperatures, and chemical concentrations—are structurally bounded and must remain strictly positive ($X(t) > 0$). In continuous-time models like Geometric Brownian Motion, this property is naturally preserved

because the drift and diffusion coefficients scale proportionally with the current state, causing the boundary at zero to be mathematically unreachable.

However, the linear discrete-time Euler-Maruyama update rule can violate this constraint:

$$X_{n+1} = X_n \left(1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_n \right) \quad (38)$$

If the system experiences a sufficiently large negative random shock such that $Z_n < -\frac{1+\mu\Delta t}{\sigma\sqrt{\Delta t}}$, the multiplier becomes negative, generating a non-physical negative state variable ($X_{n+1} < 0$). This boundary breakdown is especially severe when the volatility σ is high or when the step size Δt is large.

To resolve this issue and guarantee path stability, practitioners apply Itô's Lemma to simulate the log-transformed state variable $Y(t) = \ln(X(t))$ rather than updating the linear process directly. As derived in Section 3, the transformed SDE becomes drift-adjusted and independent of the state variable's absolute magnitude:

$$dY(t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW(t) \quad (39)$$

Discretizing this log-transformed process via the Euler-Maruyama scheme yields:

$$Y_{n+1} = Y_n + \left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_n \quad (40)$$

The simulated state variable is then recovered at each step by exponentiating the result:

$$X_{n+1} = \exp(Y_{n+1}) \quad (41)$$

Because the exponential function maps the real line onto the strictly positive domain ($\exp(Y) > 0, \forall Y \in \mathbb{R}$), this transformation prevents the system from generating negative values, ensuring numerical and physical stability across all simulated trajectories.

5 Applications in Computational Finance

5.1 Geometric Brownian Motion and the Black-Scholes Framework

In quantitative finance, asset prices cannot be modeled using a standard linear SDE because prices are scale-invariant and must remain strictly positive. To satisfy these conditions, the price of a financial asset $S(t)$ is modeled as a Geometric Brownian Motion (GBM), which is governed by the non-linear stochastic differential equation:

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t) \quad (42)$$

where $\mu \in \mathbb{R}$ represents the constant expected rate of return (drift), $\sigma > 0$ denotes the market volatility (diffusion), and $W(t)$ is a standard Wiener process.

To solve this SDE analytically, we define the log-price transformation $f(S) = \ln(S)$. Computing the partial derivatives yields:

$$\frac{\partial f}{\partial S} = \frac{1}{S}, \quad \frac{\partial^2 f}{\partial S^2} = -\frac{1}{S^2}, \quad \frac{\partial f}{\partial t} = 0 \quad (43)$$

Applying Itô's Second Lemma (derived in Section 3) to the process $f(S(t))$ gives:

$$d(\ln(S)) = \left(0 + \mu S \cdot \frac{1}{S} + \frac{1}{2}\sigma^2 S^2 \cdot \left(-\frac{1}{S^2} \right) \right) dt + \sigma S \cdot \frac{1}{S} dW(t) \quad (44)$$

$$= \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW(t) \quad (45)$$

Because the right side of this expression is independent of S , we can integrate both sides directly from 0 to T :

$$\ln(S(T)) - \ln(S(0)) = \left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W(T) \quad (46)$$

Exponentiating both sides yields the exact analytical solution for Geometric Brownian Motion:

$$S(T) = S(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma W(T)\right) \quad (47)$$

5.2 Derivation of the Black-Scholes Option Pricing Formula

A European call option gives the holder the right, but not the obligation, to buy a share of stock at a fixed maturity date T for a specified strike price K . Under the risk-neutral pricing measure, the drift μ is replaced by the risk-free interest rate r . The fair value of the option today, C_0 , is given by the discounted expected value of its terminal payoff:

$$C_0 = e^{-rT} \mathbb{E}[\max(S(T) - K, 0)] = e^{-rT} \int_K^\infty (s - K) f_{S(T)}(s) ds \quad (48)$$

where $f_{S(T)}(s)$ is the log-normal probability density function of the terminal stock price. This expectation can be split into two separate integrals:

$$C_0 = e^{-rT} \int_K^\infty s f_{S(T)}(s) ds - e^{-rT} K \int_K^\infty f_{S(T)}(s) ds \quad (49)$$

Evaluating the second integral represents the probability that the option expires in-the-money under the risk-neutral measure:

$$e^{-rT} K \int_K^\infty f_{S(T)}(s) ds = e^{-rT} K N(d_2) \quad (50)$$

where $N(\cdot)$ is the cumulative distribution function of a standard normal random variable, and:

$$d_2 = \frac{\ln(S(0)/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \quad (51)$$

Evaluating the first integral by completing the square in the exponent handles the asset-weighted probability distribution, yielding $S(0)N(d_1)$. Combining these terms results in the definitive Black-Scholes pricing formula:

$$C_0 = S(0)N(d_1) - Ke^{-rT}N(d_2) \quad (52)$$

where the upper integration limit index d_1 is defined as:

$$d_1 = \frac{\ln(S(0)/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = d_2 + \sigma\sqrt{T} \quad (53)$$

5.3 Algorithmic Verification via Monte Carlo Integration

While the Black-Scholes formula provides an exact closed-form solution for simple European options, numerical approximation methods are essential for pricing path-dependent options (such as Asian or barrier contracts) that lack analytical solutions. Monte Carlo simulation approximates the expected value directly by generating a large number of independent sample paths.

To implement this algorithm, we simulate M independent asset price trajectories over N discrete time steps using the Euler-Maruyama discretization scheme:

$$S_{n+1}^{(j)} = S_n^{(j)} + rS_n^{(j)}\Delta t + \sigma S_n^{(j)}\sqrt{\Delta t}Z_n^{(j)}, \quad \forall j \in \{1, 2, \dots, M\} \quad (54)$$

where $Z_n^{(j)} \sim \mathcal{N}(0, 1)$ are independent standard normal random variables. Let $S^{(j)}(T)$ denote the simulated terminal asset price for path j . The fair option value is estimated by averaging the discounted terminal payoffs across all simulated paths:

$$C_{\text{MC}} \approx e^{-rT} \frac{1}{M} \sum_{j=1}^M \max \left(S^{(j)}(T) - K, 0 \right) \quad (55)$$

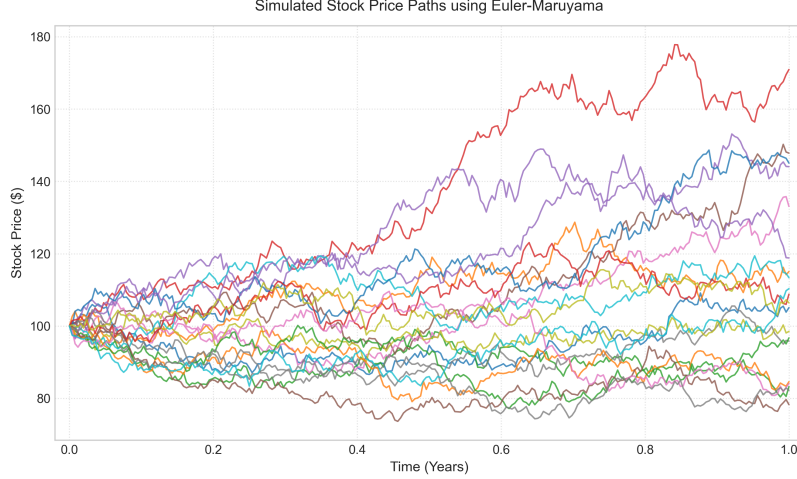


Figure 3: Simulated asset price pathways generated using the Euler-Maruyama scheme ($M = 20$, $N = 252$). The paths illustrate how stochastic diffusion wraps around a deterministic upward trend.

6 Mean-Reverting Systems (The Ornstein-Uhlenbeck Process)

6.1 Analytical Solution via Integrating Factors

While Geometric Brownian Motion models explosive, scale-invariant asset dynamics, many physical and economic variables—such as interest rates, commodity prices, and atmospheric temperatures—exhibit a natural tendency to return to a long-term average over time. The standard framework used to model this behavior is the Ornstein-Uhlenbeck (OU) process, which is governed by the linear stochastic differential equation:

$$dX(t) = \theta(\mu - X(t)) dt + \sigma dW(t) \quad (56)$$

where $\mu \in \mathbb{R}$ represents the long-term equilibrium mean, $\theta > 0$ denotes the speed of mean reversion, $\sigma > 0$ is the volatility parameter, and $W(t)$ is a standard Wiener process.

To solve this SDE analytically, we rearrange the deterministic drift components:

$$dX(t) + \theta X(t) dt = \theta \mu dt + \sigma dW(t) \quad (57)$$

We introduce a deterministic integrating factor $e^{\theta t}$. Applying the standard product rule to the combined continuous function $d(e^{\theta t} X(t))$ yields:

$$d \left(e^{\theta t} X(t) \right) = e^{\theta t} dX(t) + \theta e^{\theta t} X(t) dt \quad (58)$$

Substituting our rearranged SDE directly into this relation simplifies the expression:

$$d \left(e^{\theta t} X(t) \right) = \theta \mu e^{\theta t} dt + \sigma e^{\theta t} dW(t) \quad (59)$$

Integrating both sides of this equation from 0 to t gives:

$$e^{\theta t} X(t) - X(0) = \int_0^t \theta \mu e^{\theta s} ds + \int_0^t \sigma e^{\theta s} dW(s) \quad (60)$$

Evaluating the deterministic integral on the right side simplifies to $\mu(e^{\theta t} - 1)$. Isolating $X(t)$ by multiplying the entire equation by $e^{-\theta t}$ yields the exact analytical solution for the Ornstein-Uhlenbeck process:

$$X(t) = \mu + e^{-\theta t}(X(0) - \mu) + \sigma e^{-\theta t} \int_0^t e^{\theta s} dW(s) \quad (61)$$

6.2 Derivation of Statistical Moments

Because the Itô stochastic integral has an expected value of zero ($\mathbb{E}[\int_0^t e^{\theta s} dW(s)] = 0$), taking the expectation of the analytical solution directly gives the first statistical moment:

$$\mathbb{E}[X(t)] = \mu + e^{-\theta t}(X(0) - \mu) \quad (62)$$

Evaluating the long-time limit as $t \rightarrow \infty$ shows that the expectation converges exponentially toward the equilibrium mean, independent of the starting state:

$$\lim_{t \rightarrow \infty} \mathbb{E}[X(t)] = \mu \quad (63)$$

To derive the variance, we isolate the random fluctuations around the expected path and apply the Itô Isometry property ($\mathbb{E}[(\int_0^t f(s) dW(s))^2] = \int_0^t f(s)^2 ds$):

$$\text{Var}(X(t)) = \mathbb{E} \left[\left(\sigma e^{-\theta t} \int_0^t e^{\theta s} dW(s) \right)^2 \right] \quad (64)$$

$$= \sigma^2 e^{-2\theta t} \int_0^t e^{2\theta s} ds \quad (65)$$

$$= \sigma^2 e^{-2\theta t} \left[\frac{e^{2\theta t} - 1}{2\theta} \right] \quad (66)$$

Simplifying this expression yields the time-dependent variance formula:

$$\text{Var}(X(t)) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta t}) \quad (67)$$

Evaluating the long-time limit as $t \rightarrow \infty$ reveals that the variance stabilizes at a constant value, confirming the existence of a stationary probability distribution:

$$\lim_{t \rightarrow \infty} \text{Var}(X(t)) = \frac{\sigma^2}{2\theta} \quad (68)$$

6.3 Numerical Discretization and Simulation Analysis

To simulate the Ornstein-Uhlenbeck process, we apply the Euler-Maruyama discretization scheme over a uniform grid with step size Δt :

$$X_{n+1} = X_n + \theta(\mu - X_n)\Delta t + \sigma\sqrt{\Delta t}Z_n, \quad Z_n \sim \mathcal{N}(0, 1) \quad (69)$$

The drift term $\theta(\mu - X_n)\Delta t$ acts as an algorithmic rubber band: if the current state X_n is pulled far above the long-term mean μ , the drift coefficient becomes negative, pulling the next step back down. Conversely, if X_n drops below μ , the drift becomes positive, forcing the path back up.

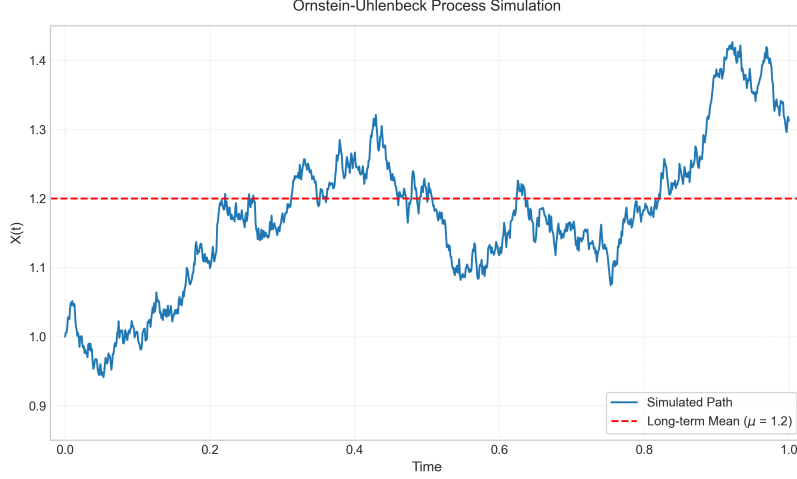


Figure 4: Numerical simulation of the Ornstein-Uhlenbeck process ($X(0) = 1.0, \mu = 1.2, \theta = 5.0, \sigma = 0.3$). The simulated path highlights continuous fluctuations around the long-term equilibrium mean.

7 Stochastic Queuing Systems and Resource Allocation

7.1 System Setup and Balance Equations

While previous sections focus on continuous-time path dynamics driven by Gaussian noise, many operational structures depend on discrete event tracking. We model an operations environment—such as a hotel reception counter—using an $M/M/1$ Markovian queue framework. The system features a single receptionist serving arriving customers on a first-in, first-out (FIFO) basis. Let the structural operational variables be defined as follows:

- λ : Mean customer arrival rate (average arrivals per unit time).
- μ : Mean customer service rate (average departures processed per unit time).
- ρ : Traffic intensity or utilization factor, defined as the ratio $\rho = \frac{\lambda}{\mu}$.
- n : Discrete state variable tracking the total number of customers currently inside the system.
- P_n : Steady-state probability that there are exactly n customers inside the system.

For the system to reach an equilibrium steady state without growing infinitely large, we enforce the stability constraint $\rho < 1$. The state configuration transitions are modeled as a continuous-time birth-death process. Under steady-state conditions, the probability flux entering any state must balance the flux leaving it. This leads to the fundamental balance equations:

$$\lambda P_{n-1} = \mu P_n \implies P_n = \left(\frac{\lambda}{\mu}\right) P_{n-1} = \rho P_{n-1} \quad (70)$$

Applying this relation recursively maps the probability of any state n directly to the empty state probability P_0 :

$$P_n = \rho^n P_0, \quad \forall n \in \{0, 1, 2, \dots\} \quad (71)$$

7.2 Derivation of Steady-State Distribution Profiles

To calculate the absolute probability scale, we invoke the total probability axiom, which states that the sum of all state probabilities must equal 1:

$$\sum_{n=0}^{\infty} P_n = 1 \implies P_0 \sum_{n=0}^{\infty} \rho^n = 1 \quad (72)$$

Because $\rho < 1$, the summation forms a convergent geometric series that simplifies to $\sum_{n=0}^{\infty} \rho^n = \frac{1}{1-\rho}$. Substituting this back into the normalization condition yields:

$$P_0 \cdot \left(\frac{1}{1-\rho} \right) = 1 \implies P_0 = 1 - \rho \quad (73)$$

Combining this result with our recursive relation provides the definitive geometric probability distribution profile for the $M/M/1$ queue length:

$$P_n = (1 - \rho)\rho^n \quad (74)$$

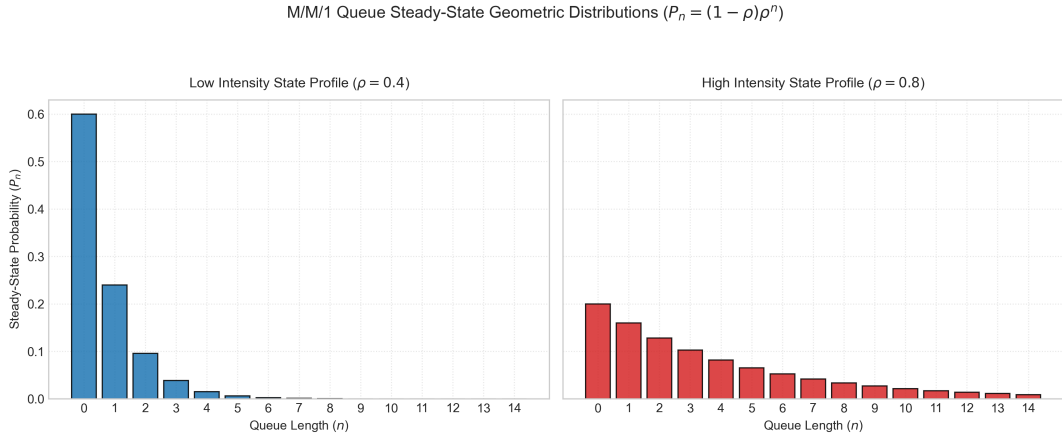


Figure 5: M/M/1 Queue Steady-State Geometric Distributions. Comparative discrete distribution maps highlighting geometric probability decay across expanding state indexes under localized load parameters (ρ).

7.3 Analysis of Performance Metrics and Waiting Tolerances

We derive the operational metrics used to evaluate resource allocation efficiency. The first metric is the average queue length L , which is defined as the expected value of the discrete state variable n :

$$L = \mathbb{E}[n] = \sum_{n=0}^{\infty} nP_n = (1 - \rho) \sum_{n=0}^{\infty} n\rho^n \quad (75)$$

Using the standard power series derivative identity $\sum_{n=0}^{\infty} n\rho^n = \frac{\rho}{(1-\rho)^2}$, the expectation evaluates cleanly to:

$$L = (1 - \rho) \cdot \frac{\rho}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} \quad (76)$$

Substituting the parameter definitions into this result gives the average number of customers inside the system as a function of the arrival and service rates:

$$L = \frac{\lambda}{\mu - \lambda} \quad (77)$$

To calculate the average time a customer spends waiting in line before their service begins (W_q), we find the expected number of people strictly waiting in the queue, denoted as L_{queue} . This is equal to the total system length L minus the expected number of customers currently being served (ρ):

$$L_{\text{queue}} = L - \rho = \frac{\rho}{1 - \rho} - \rho = \frac{\rho^2}{1 - \rho} \quad (78)$$

Finally, we apply Little's Law ($L_{\text{queue}} = \lambda W_q$) to convert this queue length into a temporal waiting metric:

$$W_q = \frac{L_{\text{queue}}}{\lambda} = \frac{1}{\lambda} \cdot \frac{\rho^2}{1 - \rho} \quad (79)$$

Substituting $\rho = \frac{\lambda}{\mu}$ yields the final waiting time formulation, illustrating how congestion increases non-linearly as traffic intensity approaches 1:

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} \quad (80)$$

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